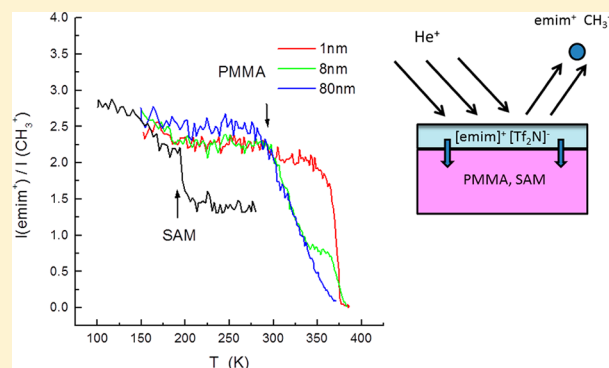


Nanoconfinement of Ionic Liquid and Polymers in Supported Thin Films: A Time-of-Flight Secondary Ion Mass Spectrometry Study of Surface Diffusion

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ABSTRACT: To gain insights into properties of a room-temperature ionic liquid in confined geometry, 1-ethyl-3-methylimidazolium bis[trifluoromethanesulfonyl]imide ([emim][Tf₂N]) adspecies interacting with various substrates was investigated as a function of temperature by using time-of-flight secondary ion mass spectrometry. The monolayer of adspecies is incorporated into substrates of porous Si, self-assembled monolayers, and Li[Tf₂N] at 180–190 K that is slightly greater than the bulk glass transition temperature, T_g , of [emim][Tf₂N]. In contrast to simple glass formers, no surface diffusivity is observed for [emim][Tf₂N] in the sub- T_g region. The uptake of [emim][Tf₂N] into polymer films is associated with the surface mobility of polymer segments rather than adspecies. The mobility of poly(methyl methacrylate) is not quenched by hydrogen bonds with the substrate, as evidenced by that the uptake onset of [emim][Tf₂N] is almost independent of the polymer film thickness.



1. INTRODUCTION

Understanding the interaction of thin liquid films with solid substrates is of fundamental and technological importance in terms of adhesion and lubrication. The physical properties of liquid in confined geometry may differ from those in the bulk because of the influence of free surface and substrate interface. In this respect, much attention has been focused on changes in glass transition temperature, T_g , of thin polymer films with decreasing thickness.^{1–15} Keddie et al.¹ first reported T_g depression of ultrathin films of polystyrene (PS) by measuring temperature evolutions of the refractivity index or film thickness using ellipsometry. They suggested that the decrease in T_g was caused by the existence of a liquidlike layer on the free surface. Keddie et al. also revealed the effects of the interaction with the substrate from the observation of increase or decrease of T_g for supported poly(methyl methacrylate) (PMMA) films.² On Si wafers with the native oxide layer, the measured T_g value was increased with decreasing film thickness whereas a decrease in T_g was observed for PMMA on gold-coated Si surface. Following these reports, interfacial effects on T_g modification have been investigated extensively for many different systems using a variety of experimental techniques.^{3–15} The experimental results generally show a dependence of the glass transition on the film thickness below 100 nm. However, contradictory results have often been encountered even for the same polymer/substrate combinations; the discrepancies could be attributed to the surface sensitivity of the experimental techniques.¹⁵ Consequently, it is not easy to assign the surface and interfacial effects on T_g modification

straightforwardly. This is because polymer chains are entangled over considerably deep layers.

On the other hand, surface mobility of simple glass formers has been explored in terms of modification of thin films' T_g . Cowin and co-workers¹⁶ probed fluidity of nanoscale domains of vapor-deposited 3-methylpentane films by measuring movements of soft-landed ions. It was revealed that ions move near the surface at temperatures much lower than the bulk T_g value, whereas the ion mobility is depressed near the substrate interface. We have explored the glass-transition behaviors of thin films by using time-of-flight secondary ion mass spectrometry (TOF-SIMS).^{17–25} It was found that the onset of thin film dewetting depends strongly on the nature of substrates: sub- T_g dewetting is observed for toluene,¹⁹ *n*-pentane,²⁰ and 3-methylpentane²⁰ films deposited onto substrates with low surface energy provided that the films are thinner than 4–5 monolayers (MLs). The enhanced surface mobility in the sub- T_g region has also been clarified by measuring uptake behaviors of thin films of water, methanol, ethanol, and 3-methylpentane into porous media.^{21,22} Thus, the effects of the free surface and substrate interface on T_g modification of nanoconfined films have been clarified at the molecular level.

In contrast to polymers and simple glass formers, very few studies have been performed concerning the nanoconfinement

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effects of ionic liquids^{26–28} despite that their bulk properties are thoroughly investigated. It was reported that more than 10-fold decrease of diffusion coefficients occurs for ionic liquids confined in oxidized porous silica membranes.²⁸ We have revealed that crystallization kinetics of 1-ethyl-3-methylimidazolium bis[trifluoromethanesulfonyl]imide ([emim][Tf₂N]) is strongly influenced by substrates for the films thinner than 2–3 ML.^{24,29} However, there exist no systematic studies on T_g modification of nanoconfined ionic liquid films or the influence of substrates on diffusivity of adspecies. On the other hand, the interactions of [emim][Tf₂N] with glassy and semicrystalline polymer films have been examined using temperature-programmed TOF-SIMS,²⁴ where ionic liquid adspecies is used as a tracer of the glass transition of the free polymer surface. [emim][Tf₂N] can be deposited thermally without decomposition and stays on the surface until higher temperatures (~ 400 K).³⁰ It was revealed that surface mobility of PS and PMMA films occurs in the sub- T_g region,²⁴ but how T_g of polymers in the monolayer regime is modified by interactions with substrates still remains unresolved.

The purpose of this paper is to discuss the nanoconfinement effects on T_g modification of [emim][Tf₂N] in comparison with the results of simple glass formers. To this end, the uptake behaviors of [emim][Tf₂N] adspecies into substrates of porous silicon, self-assembled monolayers (SAMs), Li[Tf₂N], and some polymers are explored using TOF-SIMS at cryogenic temperatures. The results of SAMs are compared with those of polymers to shed light on the influence of chain length and structure on uptake behaviors of the [emim][Tf₂N] adspecies. The effect of a hydroxylated SiO₂ substrate on surface mobility of the PMMA film is also explored based on the interaction with the [emim][Tf₂N] adspecies.

2. EXPERIMENTAL SECTION

A Si(100) wafer (p-type, 10 Ω -cm) with a native oxide layer was degreased ultrasonically in ethanol and then irradiated with a low-pressure mercury vapor lamp in air (the UV-ozone treatment) for 1 h to remove hydrocarbon contaminants. A porous silicon substrate was prepared by electrochemical etching of a Si(100) wafer (p-type, 0.01 Ω -cm).^{31,32} It was anodized at current density of 100 mA/cm² for 3 min using an electrolyte of a 1:1 mixture (by volume) of HF (55 wt %) and ethanol (99.5%). It is known that high-density mesopores with preferred growth direction perpendicular to the surface are created under this condition.³² The sample surface was hydroxylated by the UV-ozone treatment. Two types of SAMs consisting of hydrocarbon (CH₃(CH₂)₁₇-) and perfluorocarbon (CF₃(CF₂)₈-) chains, respectively, were prepared by gas-phase silanization reaction using octadecyltrimethoxysilane (ODMS, 97%, Gelest Inc.) and (heptadecafluoro-1,1,2,2-tetrahydrodecyl)trimethoxysilane (FAS, 97%, Gelest Inc.) as precursors.^{33,34} The hydroxylated Si wafers were placed in vapors of ODMS or FAS at 110 °C for 8 h. According to the literature,³⁴ the surface is coated with a homogeneous, defect free monolayer by this procedure. Thin films of PMMA (M_w = 120 000, Aldrich), poly(methyl acrylate) (PMA, M_w = 40 000, Aldrich), and poly(ethyl acrylate) (PEA, M_w = 95 000, Aldrich) were formed by spin-cast from their toluene solutions onto hydroxylated SiO₂/Si(100) wafers. The thicknesses of polymer films were determined using a profilometer (DEKTAK 3030). A thin film of Li[Tf₂N] was deposited on the SiO₂/Si(100) wafer by spin-cast from an ethanol solution.

The substrate was inserted into an UHV chamber (a base pressure of less than 1×10^{-8} Pa) via a load-lock system and was mounted on a copper coldfinger extended from a closed-cycle helium refrigerator. The sample temperature was controlled using a cartridge heater by monitoring temperature of the coldfinger close to the sample position using Au(Fe)–chromel thermocouples. The temperature was ramped at a rate of ca. 5 K min⁻¹ using a digital temperature programmer. [emim][Tf₂N] (Kanto Kagaku, 99%) was deposited onto the surface by thermal evaporation from a Ta boat placed in front of the sample surface. For TOF-SIMS measurements, a primary beam of 2 keV He⁺ ions was generated in an electron-impact-type ion source (Specs, IQE 12/38) and was chopped into pulses using a set of electrostatic deflection plates and apertures. Positive secondary ions emitted perpendicularly to the sample surface were detected using a microchannel plate after traveling a field-free TOF tube. The fluence of He⁺ was kept below 1×10^{12} ions cm⁻² to reduce damaging the surface.

3. RESULTS

Porous and nonporous Si substrates were heated slightly in UHV for cleaning; the formation of hydroxylated surface was confirmed based on the TOF-SIMS spectra.²⁴ The coverage of [emim][Tf₂N] was determined from evolutions of secondary ion intensities as a function of deposition time.²⁴ In this study, attention is focused on the monolayer film. Figure 1 shows

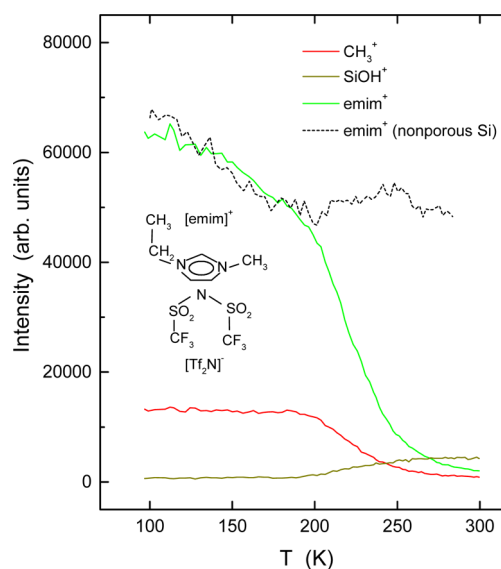


Figure 1. Temperature-programmed TOF-SIMS intensities of typical ions sputtered from the [emim][Tf₂N] monolayer deposited on a porous Si substrate that is terminated with the hydroxyl group. The result obtained using hydroxylated Si(100) surface is also shown by a broken line. The [emim][Tf₂N] adspecies is deposited at 100 K, and temperature was increased at a rate of 5 K min⁻¹. The molecular structure of [emim][Tf₂N] is displayed schematically in the inset.

temperature evolutions of typical secondary ion intensities obtained for a porous Si substrate on which [emim][Tf₂N] was deposited at 100 K. The decrease of the CH₃⁺ intensity from adspecies is observed together with the increase of the SiOH⁺ intensity from the substrate at temperatures higher than ca. 190 K. In contrast, the emim⁺ intensity decreases gradually in the plateau region of the CH₃⁺ intensity and then decreases more steeply at $T > 190$ K. To understand this behavior, the emim⁺

intensity obtained using a nonporous SiO₂/Si(100) substrate is also displayed as a broken line. The initial decrease of the emim⁺ intensity is commonly observed for both substrates. This phenomenon has been explained as reorganization of the vapor-deposited [emim][Tf₂N] molecules.²⁴ The decay of the emim⁺ intensity at $T > 190$ K results from uptake of the adspecies into mesopores, indicating that surface diffusivity of [emim][Tf₂N] occurs at this temperature. For [emim][Tf₂N] multilayers deposited on the nonporous SiO₂ substrate, it is known that the emim⁺ intensity increases significantly upon crystallization at ca. 210 K.²⁴ The crystal-like ordering is observed even for the monolayer of [emim][Tf₂N] when deposited onto highly oriented pyrolytic graphite.²⁹ However, such a behavior is absent on the SiO₂ substrate, suggesting that the adspecies–adspecies interaction is perturbed significantly by the adspecies–substrate interaction.

Figure 2 shows TOF-SIMS intensities from an ODS layer formed on the nonporous SiO₂/Si(100) substrate after

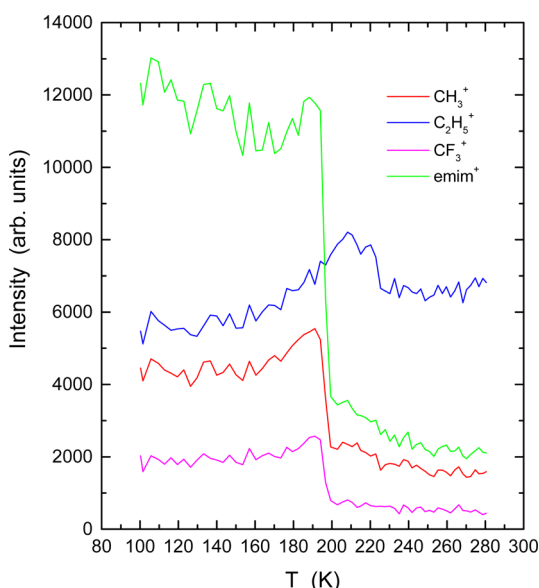


Figure 2. Temperature-programmed TOF-SIMS intensities from the [emim][Tf₂N] monolayer deposited on an ODS-terminated SiO₂ substrate.

deposition of 1 ML of [emim][Tf₂N]. The CH₃⁺ and C₂H₅⁺ ions come from both substrate and adspecies, but the CF₃⁺ ion arises only from the Tf₂N moiety of adspecies. The occurrence of surface mobility at around 195 K is clearly identifiable. The steep decay of the ion intensities compared to the result for the porous Si substrate suggests that the adspecies is incorporated more rapidly on this surface after the evolution of surface mobility. It is also possible that the morphology of the [emim][Tf₂N] monolayer changes at this temperature without incorporation in the ODS film.

The experimental result using a FAS-terminated Si is displayed in Figure 3. The CF⁺ ion comes from both adspecies and substrate, but its intensity from the neutral substrate is greater than that from the [Tf₂N][−] moiety. The mobility of [emim][Tf₂N] occurs at around 185–190 K, as revealed from the ions evolution from both adspecies and substrate. The substrate terminated with perfluorocarbon is characterized by a small surface energy; the mobile adspecies tends to form droplets, as observed for adsorption of simple molecules.²⁰ It

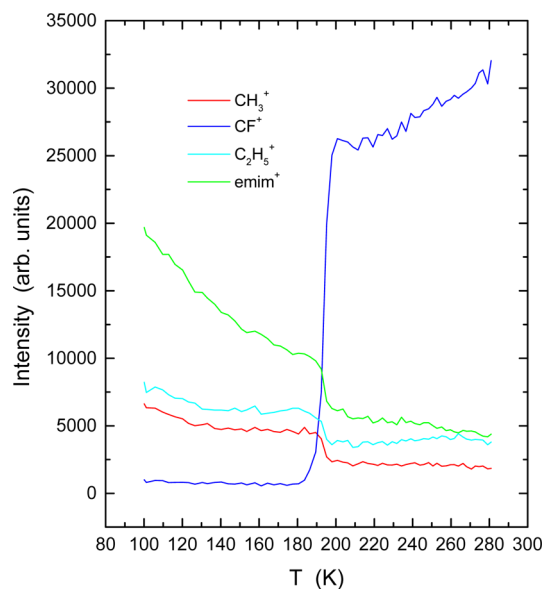


Figure 3. Same as in Figure 2, but on a FAS-terminated SiO₂ substrate.

might be presumed that the relatively high emim⁺ intensity observed after the evolution of mobility is an indication of the droplet formation. Thus, it is not clear at this stage whether the mobile [emim][Tf₂N] adspecies form droplets or permeate into the FAS layer. The situation is same for [emim][Tf₂N] on the ODS layer: the emim⁺ intensity remains at higher temperature as seen in Figure 2.

The behavior of ionic liquids on an ionic crystal surface is of interest because the cohesive force of both systems is thought to be ionic in nature. The experimental result for [emim]-[Tf₂N] deposited on a Li[Tf₂N] substrate is shown in Figure 4. The surface mobility of adspecies occurs at around 190 K, as revealed from the evolutions of CH₃⁺, C₂H₅⁺, and Li⁺ intensities, although the emim⁺ intensity decreases rather gradually across this temperature because structural relaxation occurs at lower temperatures. The wettability of adspecies is

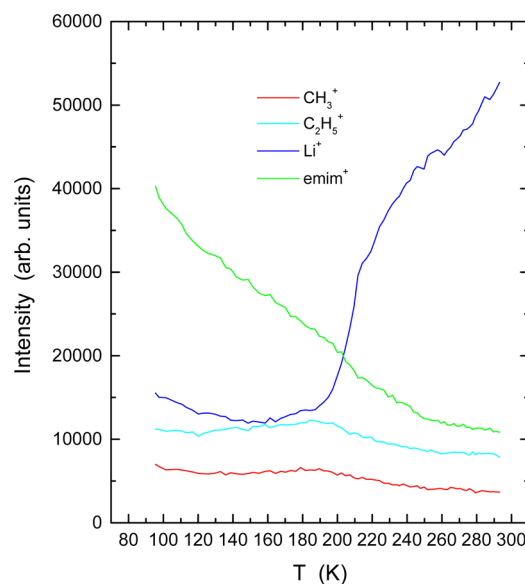


Figure 4. Temperature-programmed TOF-SIMS intensities from the [emim][Tf₂N] monolayer deposited onto a Li[Tf₂N] film.

expected to be high on this surface because of the common anion. Moreover, the cations exchange might lead to mixing of emim^+ and Li^+ ions without translational molecular diffusion. In any case, the diffusive $[\text{emim}][\text{Tf}_2\text{N}]$ adspecies cannot migrate deeper into the substrate, as evidenced by that intensities not only of the intact emim^+ cation but also of its fragment ions (CH_3^+ and C_2H_5^+) remain considerably high at higher temperatures. This result can be explained as surface segregation of emim^+ after the occurrence of mixing. In fact, it has been revealed that the surface composition of emim^+ is high for a spin-cast film of methanol solutions of dilute $[\text{emim}][\text{Tf}_2\text{N}]$ in $\text{Li}[\text{Tf}_2\text{N}]$ at room temperature because a larger ion tends to stay on the outermost surface.³⁵

Thus, an essentially common onset of 180–190 K is observed in diffusivity of 1 ML of $[\text{emim}][\text{Tf}_2\text{N}]$ on substrates of the porous Si, SAMs, and $\text{Li}[\text{Tf}_2\text{N}]$. In contrast to these results, however, it has been reported that the uptake of 2–3 ML of $[\text{emim}][\text{Tf}_2\text{N}]$ on polymers such as PMMA, PS, and poly(ethylene oxide) occurs at much higher temperatures.²⁴ The origin of this behavior is explored in the following. It was revealed that the intact emim^+ ion is sputtered intensively from adspecies, whereas the fragment CH_3^+ ions are commonly ejected from both polymers and adspecies. Therefore, we use the relative intensity of emim^+ to CH_3^+ as a measure of the uptake of the adspecies into substrates. The experimental results using PMA and PEA films (thickness of ca. 100 nm) are shown in Figure 5. The relative intensity of emim^+ decreases at

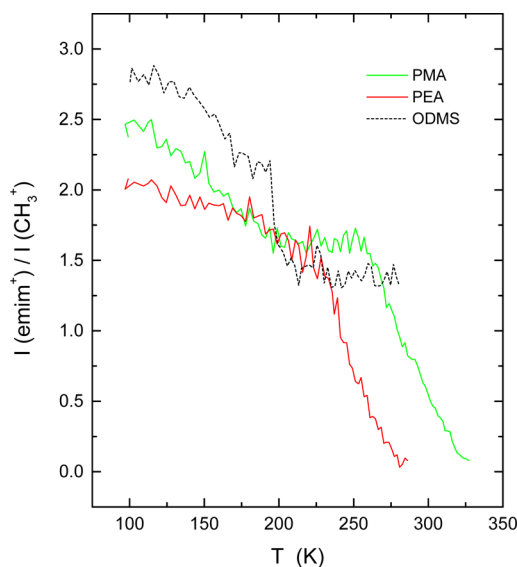


Figure 5. Intensities of emim^+ relative to CH_3^+ for the $[\text{emim}][\text{Tf}_2\text{N}]$ monolayers deposited onto PMA and PEA films. The result using an ODMS-terminated SiO_2 substrate is shown by a broken line for comparison.

temperatures of 220 and 260 K for PEA and PMA, respectively. For comparison, the result using the ODMS-terminated Si substrate is also shown by a broken line. The decay onset is strongly dependent on the polymer species and appears at higher temperature than that on ODMS. Moreover, the adspecies tends to disappear from the polymer surfaces more completely as revealed by that the final intensity of emim^+ becomes close to zero. We suggested that the presence of residues on the ODMS surface might be an indication of the droplet formation. In reality, however, this possibility can be

discarded because no indications of dewetting are recognizable at ca. 190 K on the polymer substrates whose chemical properties are expected to resemble those of ODMS. Consequently, the liquidlike monolayer of $[\text{emim}][\text{Tf}_2\text{N}]$ fundamentally wets both substrates, but its uptake behaviors are significantly different between the polymers and ODMS substrates.

The thickness dependence of polymers in interactions with adspecies is further examined using PMMA films deposited onto the hydroxylated $\text{SiO}_2/\text{Si}(100)$ substrate; the experimental results for 1 ML of $[\text{emim}][\text{Tf}_2\text{N}]$ adspecies are shown in Figure 6. The decay of the emim^+ intensity is observed at

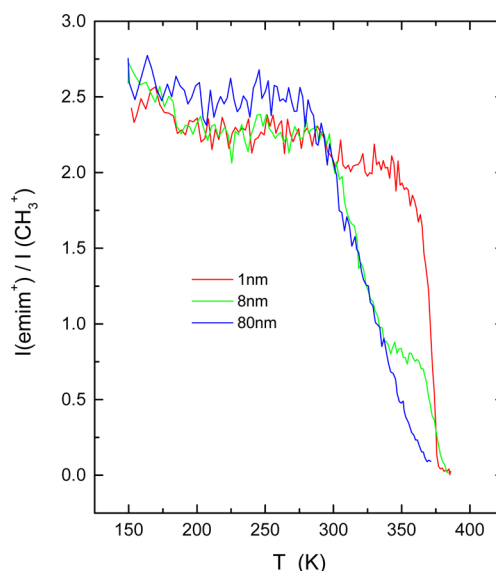


Figure 6. Relative intensities of emim^+ to CH_3^+ from the $[\text{emim}][\text{Tf}_2\text{N}]$ monolayer deposited on PMMA at 150 K. Results using PMMA films with thickness of 1, 8, and 80 nm formed on the hydroxylated SiO_2 substrate are compared.

around 300 K on the 80 nm thick PMMA film; the intensity finally drops close to zero, suggesting that almost complete uptake of adspecies results. On the 8 nm PMMA film, the uptake of adspecies starts to occur at 300 K as well, but there exists a shoulder at higher temperature because of the incomplete uptake of $[\text{emim}][\text{Tf}_2\text{N}]$; approximately 30% of the adspecies remains on the surface without permeation, as estimated from the relative intensity. In contrast, adspecies tends to stay on the surface of the 1 nm PMMA film although a slight decrease in the emim^+ intensity is identifiable at 300 K. The emim^+ intensity finally drops to zero at around 370 K because of the evaporation of adspecies. The amount of the incorporated adspecies depends on the PMMA film thickness (i.e., volume), but the uptake onset is fundamentally fixed irrespective of the film thickness. The same onset of 300 K was observed using 2–3 ML of $[\text{emim}][\text{Tf}_2\text{N}]$ adspecies on the 100 nm PMMA film.²⁴ Thus, the mobility onset of $[\text{emim}][\text{Tf}_2\text{N}]$ at 190 K is not identified on the polymer films examined here; the uptake temperature of adspecies is strongly dependent on the polymer species.

4. DISCUSSION

The surface diffusion of $[\text{emim}][\text{Tf}_2\text{N}]$ is explored based on its uptake behaviors into various substrates. The diffusivity commences on porous Si, SAMs, and $\text{Li}[\text{Tf}_2\text{N}]$ at 180–190

K, which is slightly higher than T_g of [emim][Tf₂N] (175 K).³⁶ This behavior contrasts sharply to that of simple molecules whose surface diffusivity occurs at temperatures considerably lower than T_g (e.g., $\sim 0.6 T_g$ for toluene,²⁰ ethylbenzene,¹⁹ and 3-methylpentane²⁰). The formation of a highly mobile layer near the surface is reasonable due to density suppression and disruption of molecular packing. From the point of view of the finite size effect, moreover, the cooperative motion of small domains is more facilitated than the larger one, so that mobility of the monolayer is expected to be enhanced relative to that in the bulk. On the other hand, it is demonstrated that surface diffusivity of adspecies is hampered by an attractive interaction with a solid substrate. Such a phenomenon has been observed typically for methanol and ethanol monolayers adsorbed on the hydroxylated Si surface^{21,22} because the formation of hydrogen bonds with the substrate quenches a liquidlike, cooperative motion of the adspecies.

No surface mobility of the [emim][Tf₂N] monolayer is observed in the sub- T_g region using porous silica, suggesting the role of adspecies–substrate attractive interactions. In this respect, Iacob et al.²⁸ reported that the diffusion coefficient of 1-hexyl-3-methylimidazolium hexafluorophosphate confined in oxidized nanoporous silica membranes is reduced by 10 times relative to that in the bulk based on charge transport measurements. The results are explained as the formation of hydrogen bonding between the ionic liquid and the silanole groups on pore walls. Moreover, Singh et al.²⁶ reported changes in the phase behavior of ionic liquid confined in a nanoporous silica gel matrix; ΔT_g is rather small compared to the significant changes in the melting point and crystallization temperature. The present result appears to be consistent with these observations. Iacob et al.²⁸ also found that the diffusion coefficients increase remarkably upon silanization of the silica membranes because hydrogen bonding is extinguished by termination of the surface with the CH₃ group. In the present study, however, the diffusivity onset is almost same between the porous silica and SAMs, implying that the hydrogen bond has marginal effects on adspecies–substrate interactions. The interaction of [emim][Tf₂N] with Li[Tf₂N] should be ionic, but mixing commences at around the same temperature as observed for the nonionic substrates. What was observed here concerns solvation rather than simple surface diffusion, but mobility of adspecies is prerequisite for mixing to occur with substrates. The interaction of [emim][Tf₂N] with SAMs at $T > T_g$ can also be associated with solvation. No significant difference is recognizable between aliphatic and perfluorocarbon species. In contrast to these results, the sub- T_g mobility of simple hydrocarbon molecules has been observed specifically on the substrate terminated with perfluorocarbon species, as evidenced by the formation of droplets.^{19,20} The formation of a wetting [emim][Tf₂N] monolayer is indicative of the attractive interaction with the substrates examined here. It is possible that a long-range ionic interaction of the [emim][Tf₂N] adspecies restrains the finite size effect. Probably, sub- T_g mobility of the [emim][Tf₂N] adspecies is quenched by these effects. The vapor-deposited molecular solid films are generally characterized by low density or a porous structure; surface diffusion of molecules on pore walls leads to film densification in the sub- T_g region.^{18,19} The decrease in the emim⁺ intensity at $T < 190$ K is an indication of structural relaxation of molecules,²⁴ but this behavior has nothing to do with surface diffusion of adspecies.

The uptake onset of [emim][Tf₂N] depends strongly on the polymer species. The mobile [emim][Tf₂N] adspecies is

expected to be formed on the surface at temperatures greater than 180–190 K irrespective of the substrate. Therefore, the experimental results show that the liquidlike monolayer cannot be incorporated unless the mobility of the polymer surface occurs. For PMMA, the uptake onset of ca. 300 K is lower than bulk T_g of 379 K,²⁴ indicating that a liquidlike layer is formed on the surface in the sub- T_g region. The uptake onsets on the PMA and PEA films, respectively, are also lower than their bulk T_g 's of 282 and 250 K, although these films might crystallize in part because their T_g 's are lower than the sample preparation temperature of 300 K. The mobility of polymers in the sub- T_g region may be associated with segmental motion rather than center-of-mass (CM) diffusion of polymer chains. In this respect, a comparison with the results of SAMs like ODMS is instructive. The uptake of adspecies immediately after the evolution of their surface mobility implies that the aliphatic chain of the ODMS layer is mobile up to 180–190 K although its CM diffusivity is quenched because one end of the chain is anchored to the SiO₂ substrate. It is also likely that open ends of the ODMS layer facilitate the uptake of adspecies. A smaller amount of adspecies incorporated into the ODMS and FAS layers is ascribable to the volume effect. The volume effect is also identifiable as a thickness dependence of the PMMA film (Figure 6).

In the present study, a liquidlike property is observed on the surface of the PMMA films at temperatures almost 80 K below the bulk T_g value, but no apparent film thickness dependence is identified. This behavior appears to be incompatible with the conclusion of the ellipsometry study for the same system,² where an increased T_g with decreasing film thickness (i.e., the stiffening effect) has been attributed to the formation of a “dead layer” near the interface. The dead layer might be formed as a result of the attractive interactions via hydrogen-bond formation between hydroxyl groups of the substrate surface and PMMA, thereby quenching diffusivity of polymers. In contrast to simple molecules like methanol and ethanol, the hydrogen bonds between PMMA and the substrate do not freeze the motion of long chains completely, as evidenced by the uptake in part of adspecies into the thinnest (1 nm) PMMA layer. From this context, the segmental mobility is inherent in the polymer's glass transition and is distinct from the CM mobility on which the glass transition of simple molecules relies. Since the former is likely to occur in the sub- T_g region, it can be regarded as structural relaxation or β -relaxation of polymers. In this respect, Torkelson and co-workers³⁷ discussed the surface and interface effects on structural relaxation of PMMA. From the fluorescence measurements of 35 nm PMMA films at 305 K, they observed that relaxation rate is reduced by a factor of 4 on a silica substrate relative to the bulk. The occurrence of structural relaxation for the monolayer in direct contact with an attractive substrate is fundamentally compatible with the present result. On the other hand, what was observed in the ellipsometry might concern CM diffusion (α -relaxation) of PMMA. The CM diffusion of thin liquid films can be identified as morphology change or dewetting.^{17,18} In reality, however, dewetting of liquidlike monolayers is quenched because of the attractive interaction with the substrate.^{20,21} This is also evidenced by the fact that the liquidlike [emim][Tf₂N] monolayer wets the polymer surfaces at $T > T_g$, as revealed in this study. Therefore, the so-called dead layer of PMMA is characterized as a wetting monolayer and may not necessarily be static in terms of both segmental and CM mobility. More detailed information about CM

diffusivity of nanoconfined PMMA is required for understanding the roles of segmental mobility in T_g modification of polymers.

5. CONCLUSION

Interactions of [emim][Tf₂N] adspecies with various solid substrates were investigated to gain insights into properties of ionic liquids in nanoconfined geometry. The adspecies are incorporated into porous Si, SAMs, and Li[Tf₂N] substrates at 180–190 K. No surface diffusivity of monolayer is identified in the sub- T_g region, in contrast to simple molecules. This behavior may be attributable to the presence of attractive adspecies–substrate interactions and the long-range ionic interactions. In contrast, the [emim][Tf₂N] monolayer tends to stay on the surface of polymers until temperatures much greater than T_g without dewetting or permeation. The uptake temperature depends strongly on the polymer species because the onset is associated with mobility of the polymer surface rather than adspecies. The possibility of segmental mobility without CM diffusion is suggested from comparison of the uptake behaviors between SAMs and polymer surfaces. The dead layer of PMMA is expected to be formed at the interface of the hydroxylated SiO₂ substrate because of the hydrogen-bond formation, but the surface mobility of the nanoconfined PMMA film is observed in the sub- T_g region irrespective of the film thickness.

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Notes

The authors declare no competing financial interest.

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